

1 Qubit VQLS Implementation

Goal: The goal for this implementation is to try and make the smallest VQLS possible and show a process that can solve a 2x2 matrix and understand the outputs that come from it

First I'll outline the framework that I'm working back from, my goal with this is to:

- ☐ Get a refresher on the VQLS algorithm (alr partially done by reviewing the paper)
- ☐ Understand the outputs that come from VQLS → so after I'll revisit the 3-qubit and then the 10-qubit to better understand

First I'm gonna take some random A and $|b\rangle$, so lets say that

$$A = \mathbb{I} + 0.5X + 0.5Z$$
$$|b\rangle = H|0\rangle$$

So in matrix form this means that

$$A = \begin{bmatrix} 1.5 & 0.5 \\ 0.5 & -0.5 \end{bmatrix} \text{ and } |b\rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

So the solution to this equation is:

$$|x\rangle = A^{-1}|b\rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}$$

I'm starting with trying to build this circuit without a hadamard test. Since it's only 1-qubit, it makes sense that I should just be able to calculate the expectation value directly for the cost function.

In this implementation I'm using a simple ansatz of just a hadamard gate and a single rotation gate around the y-axis.

$$|0\rangle \text{ --- } \boxed{H} \text{ --- } \boxed{R_y(\theta)} \text{ --- }$$

So only 1 parameter is being optimized in this algorithm.